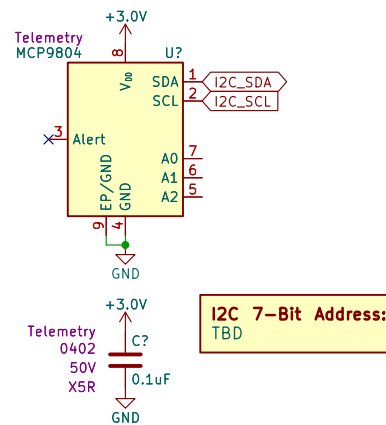
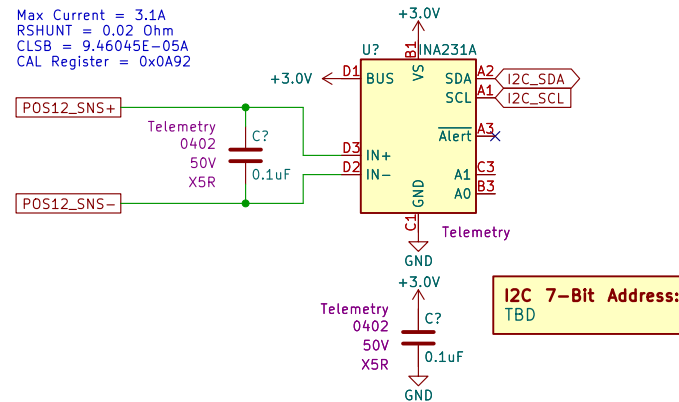


1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch LED Power Supply File: led_power_supply.kicad_sch LED Telemetry File: led_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch			A
B					B
C					C
D					D
1	2	3	4	5	



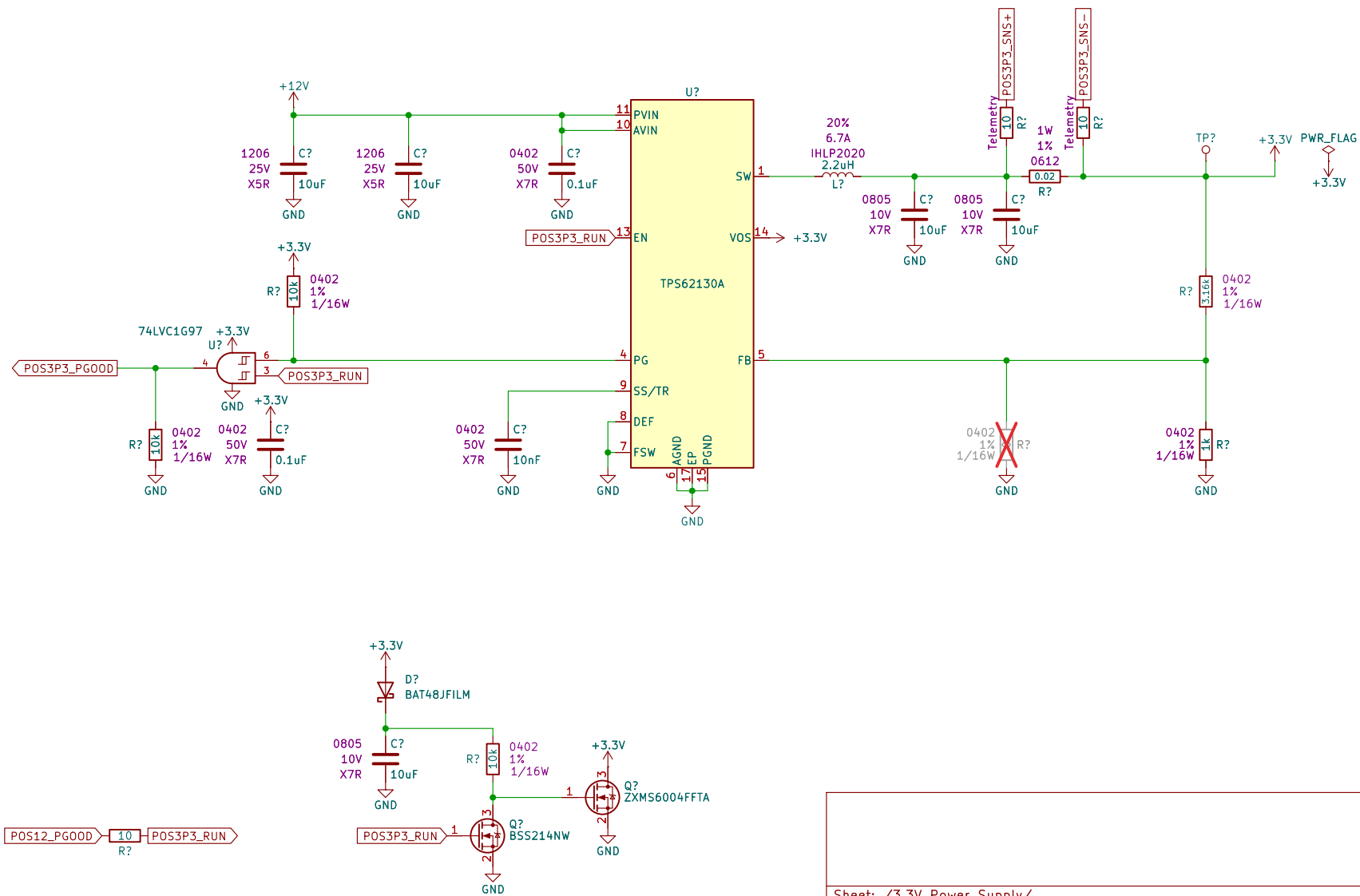
Sheet: /12V Telemetry/
 File: POS12_telemetry.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 3/19

TRY TO TALLY LOAD CURRENTS



Sheet: /3.3V Power Supply/ File: POS3P3_power_supply.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 4/19

1					2					3					4					5					
A																									A
B																									B
C																									C
D																									D
1					2					3					4					5					

Sheet: /3.3V Telemetry/
File: POS3P3_telemetry.kicad_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 5/19

Sheet: /3.3V Telemetry/ File: POS3P3_telemetry.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	
KiCad E.D.A. 10.0.3		Rev: Id: 5/19

1					2					3					4					5					
A																									A
B																									B
C																									C
D																									D
1					2					3					4					5					

Sheet: /LED Power Supply/
File: led_power_supply.kicad_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 6/19

Sheet: /LED Power Supply/ File: led_power_supply.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 6/19

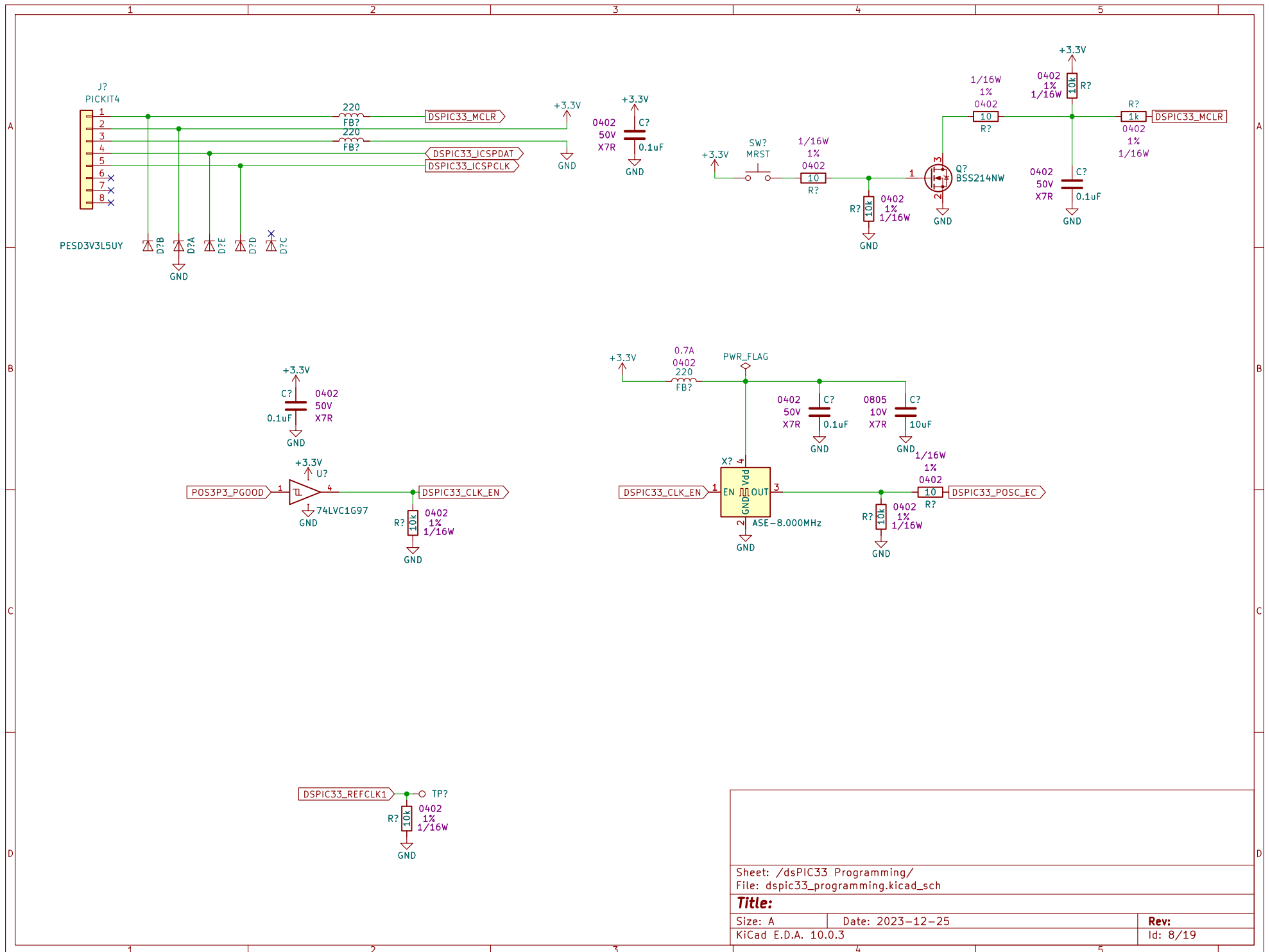
1					2					3					4					5					
A																									A
B																									B
C																									C
D																									D
1					2					3					4					5					

Sheet: /LED Telemetry/
File: led_telemetry.kicad_sch

Title:

Size: ADate: 2023-12-25

KiCad E.D.A. 10.0.3Rev: Id: 7/19

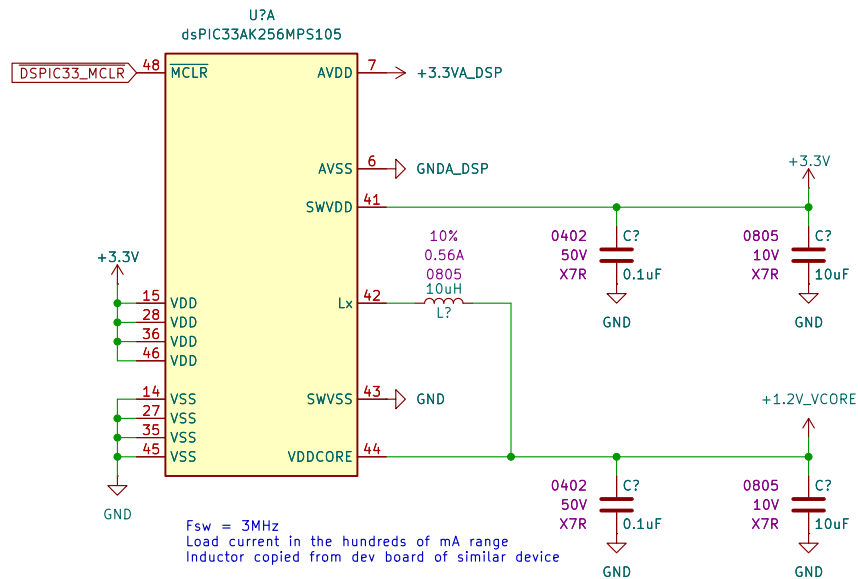
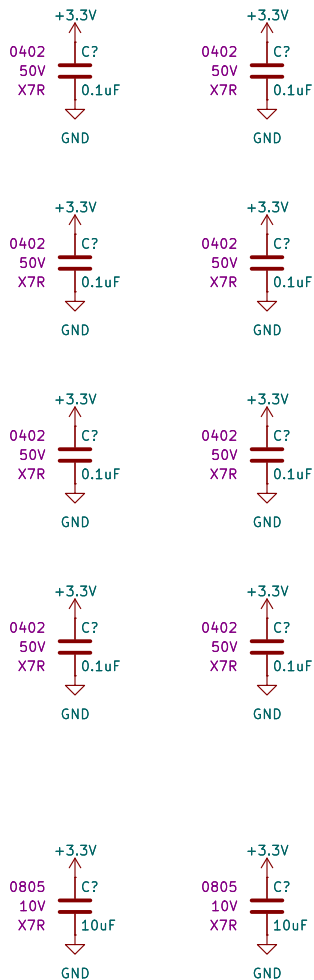


Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch

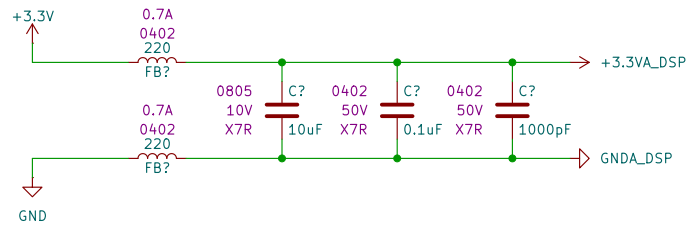
Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 8/19



Fsw = 3MHz
Load current in the hundreds of mA range
Inductor copied from dev board of similar device



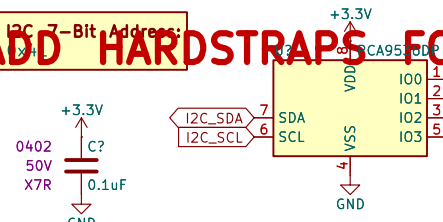
Sheet: /dsPIC33 Power/
File: dspic33_power.kicad_sch

Title:

Size: A Date: 2023-12-25

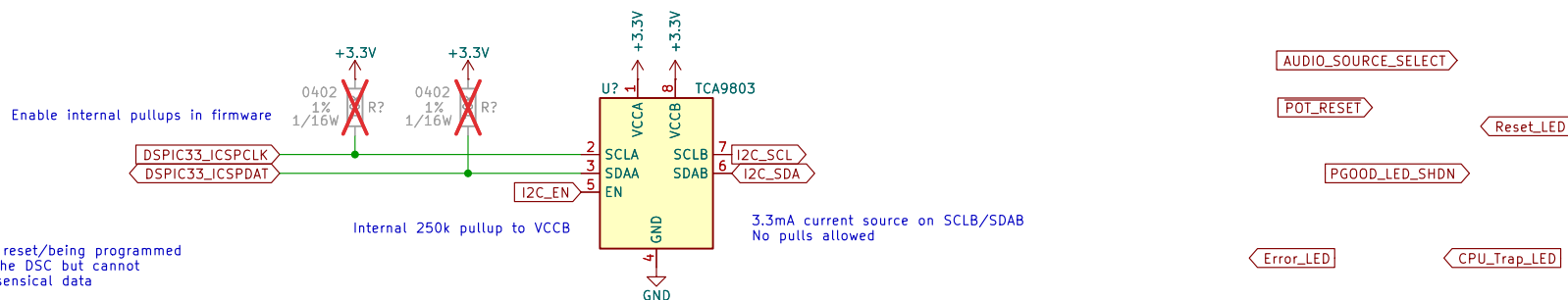
KiCad E.D.A. 10.0.3

Rev:
Id: 9/19

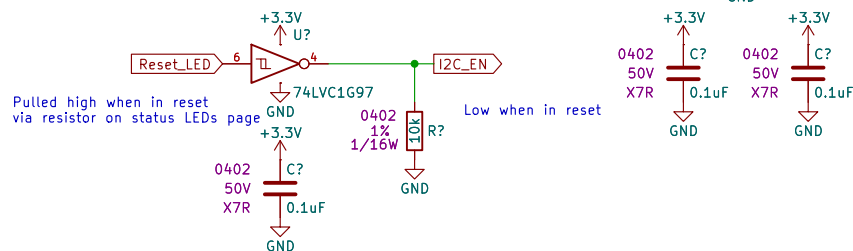


U?B

dsPIC33AK256MPS105



Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data



Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch

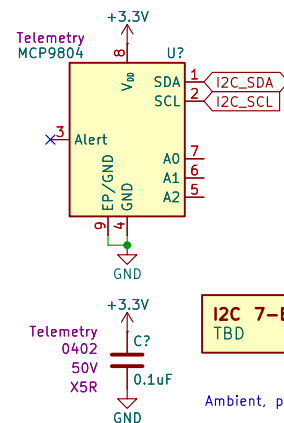
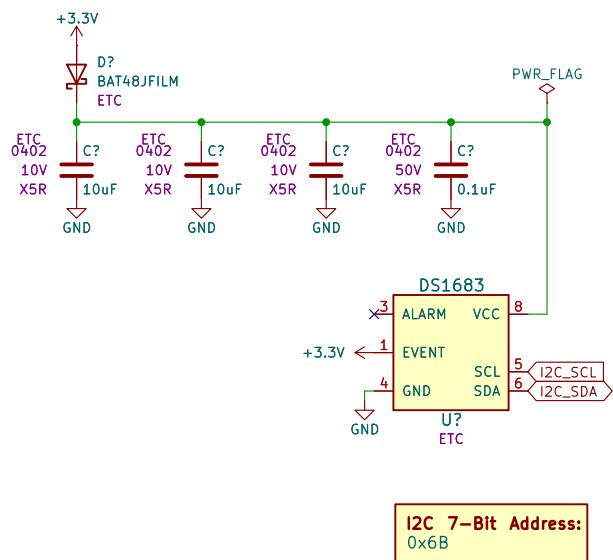
Title:

Size: A	Date: 2023-12-25
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Rev:

Id: 10/19



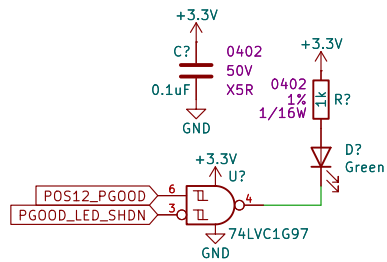
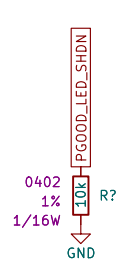
Ambient, place away from power components

Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/19



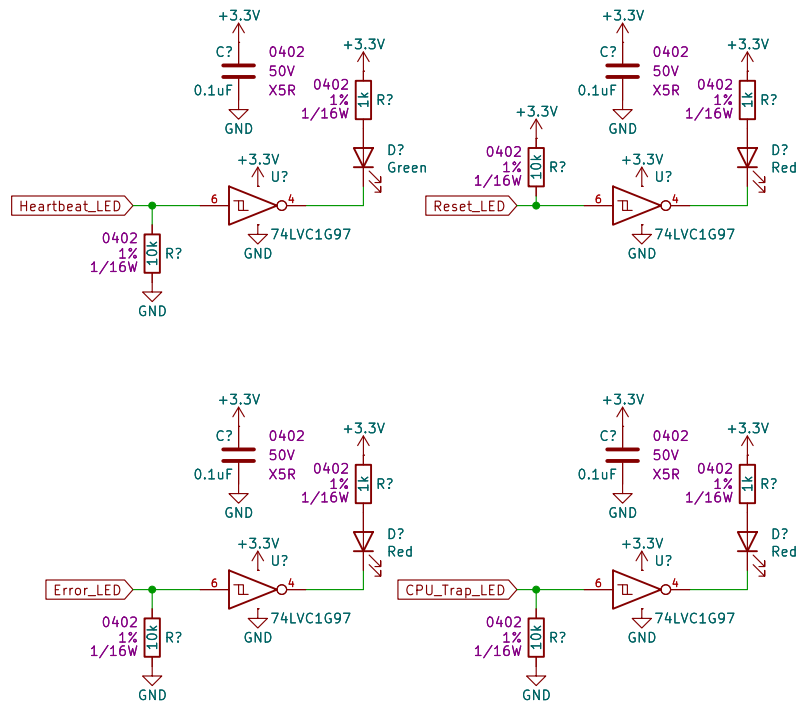
Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title:

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:
Id: 13/19

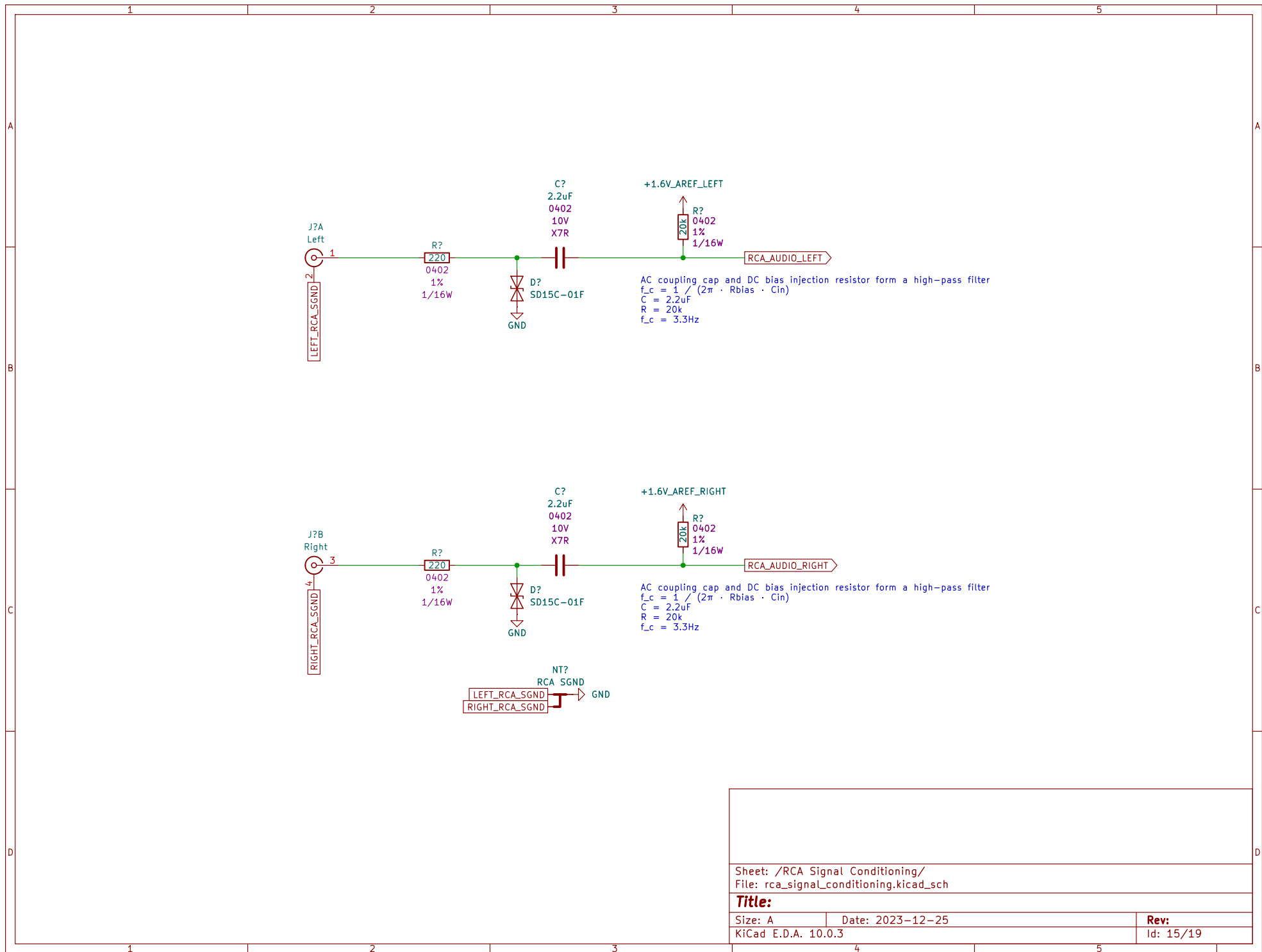


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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Id: 14/19

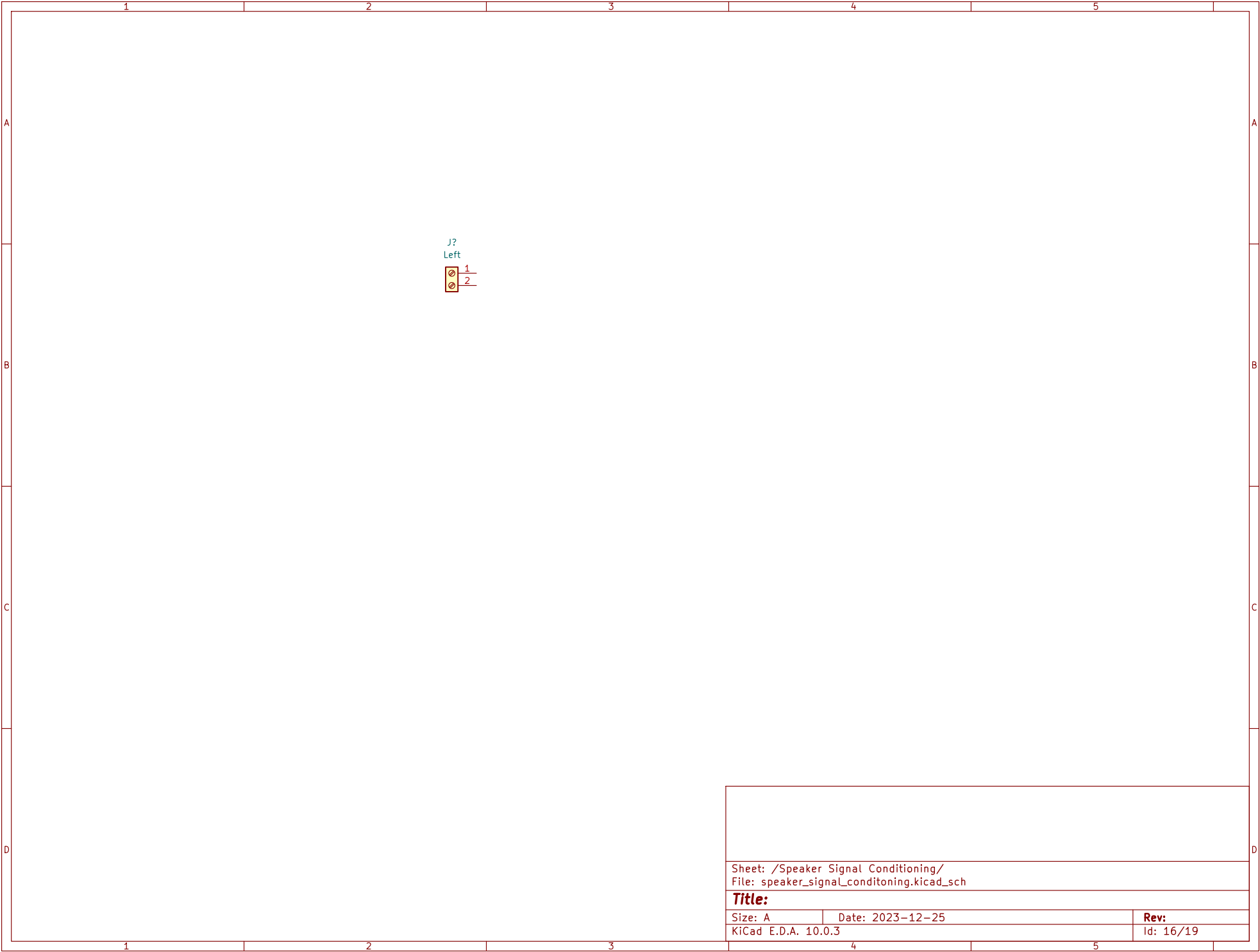


Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

Title:

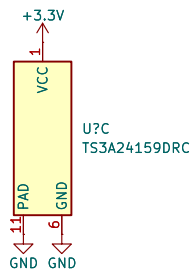
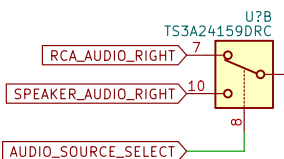
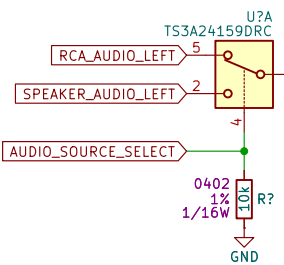
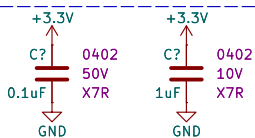
Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 15/19

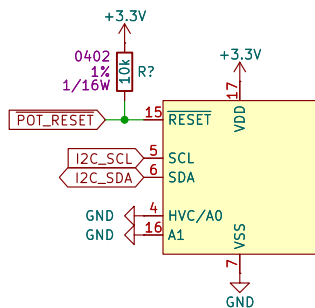
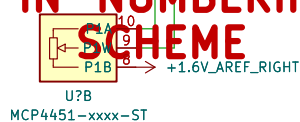
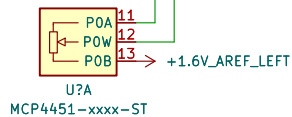


Sheet: /Speaker Signal Conditioning/ File: speaker_signal_conditoning.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 16/19

**THIS IS OFFERED
IN A QFN BUT MIGHT
HAVE A DIFFERNT
PIN NUMBERING
SCHEME**

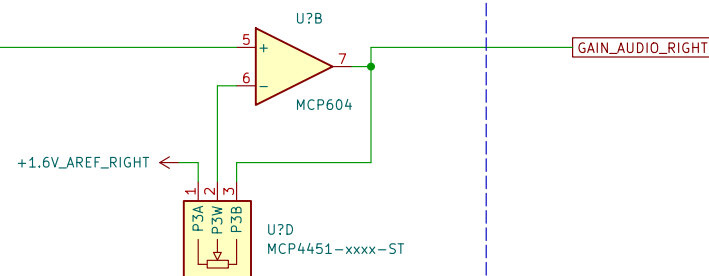
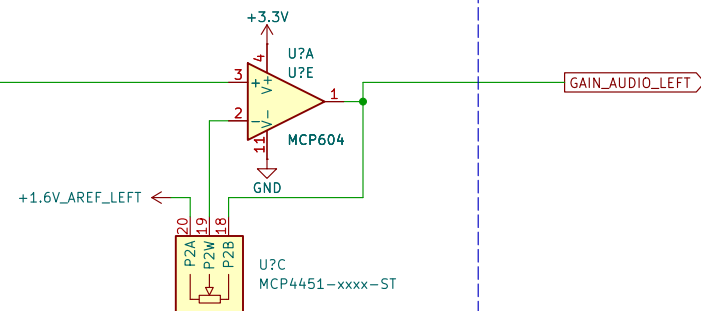
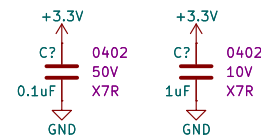


Audio source selection analog switches

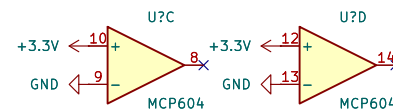


**I2C 7-Bit Address:
0x2C**

Digital potentiometers for adjustable attenuation



Digital potentiometers/non-inverting amp
for adjustable gain

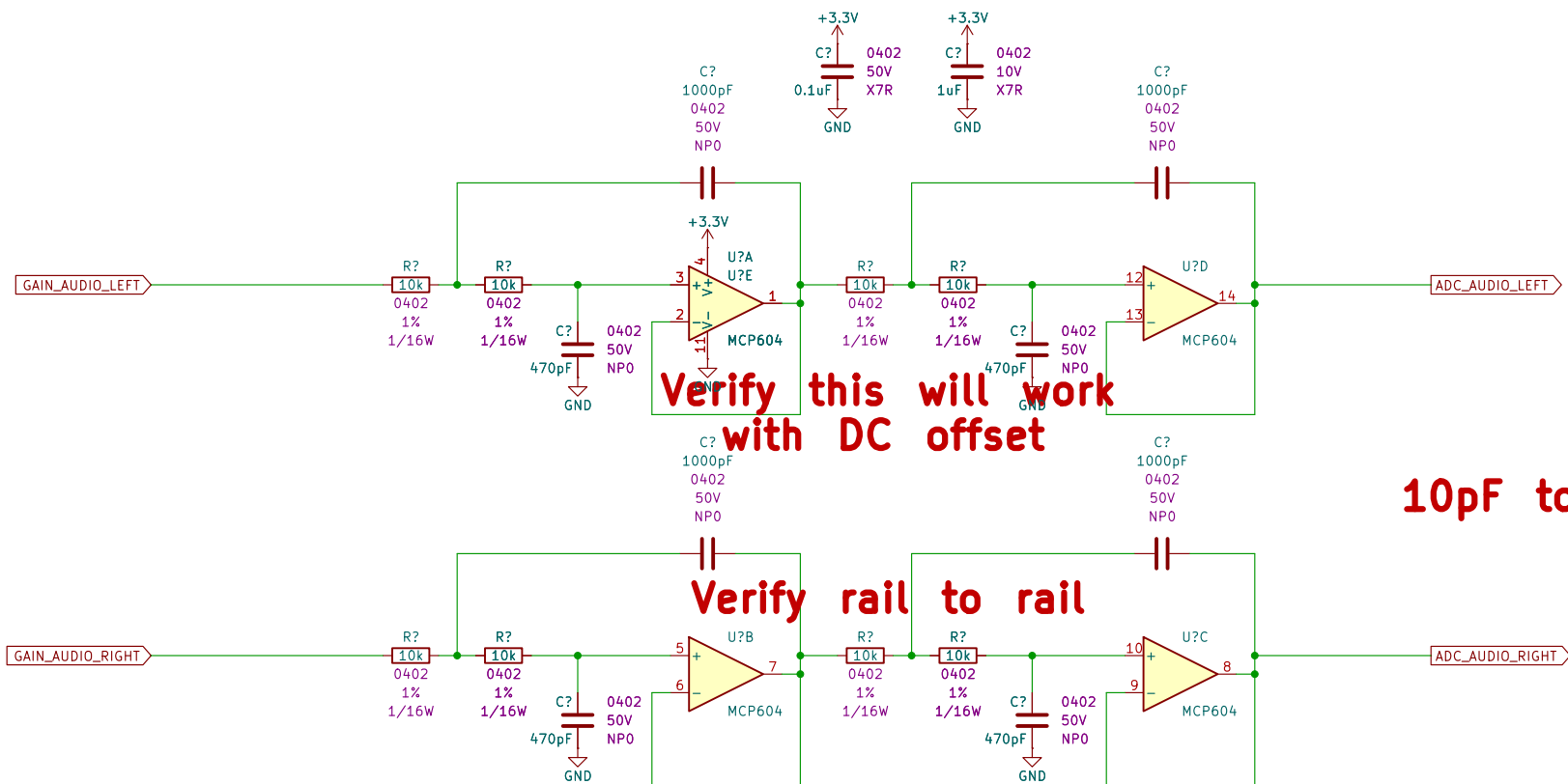


Sheet: /Source Selection, PGAs/
File: source_selection_pgas.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/19



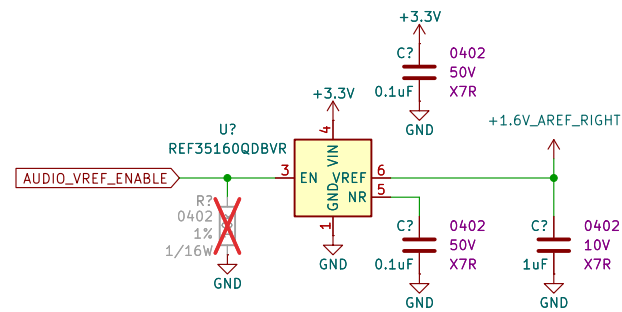
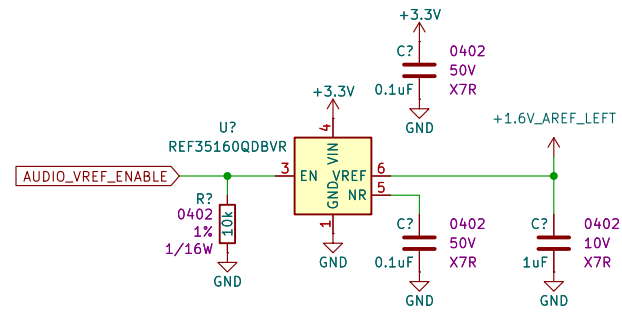
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 18/19



Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 19/19